CHE 330 Exam 1 Study Guide

The first CHE 330 exam will be a 50 minute open-book, open-note exam covering the application of statistics to engineering problems. In order to do well on this exam, you should be able to do the following:

Error Propagation in Engineering Work

1. Identify variables for which measurement uncertainty is a concern.
2. Complete error propagation analysis for an equation or series of equations by applying the following relationships:
    a. Linear Combination Relationship
    b. Power Law Relationship
3. Identify which measured variables contribute most to uncertainty in results.

Discrete Variables and Probability

1. Identify whether a given variable is considered discrete and describe why.
2. Determine probability distributions for discrete variables.
3. Calculate the expectation value, E(x), for a discrete variable and/or function of a discrete variable.
4. Calculate mean, variance (σ^2) and standard deviation (σ) for a given sample or population of discrete variables.
5. Identify whether a discrete variable has a binomial distribution and describe why.
6. Determine probability for selected binomial variables using appropriate equation(s).
7. Identify when discrete variable data can be considered to adhere to a Poisson distribution.
8. Apply Poisson distribution equation(s) to solve for probability when appropriate.

Continuous Variables and Probability

1. Identify whether a given variable is considered continuous and describe why.
2. Calculate mean, variance (σ^2) and standard deviation (σ) for a given sample or population of continuous variables.
3. Describe and apply criteria determining if it is necessary to use Z-statistics or t-statistics to solve for probability of normally-distributed variables.
4. Utilize the standard Z-table to solve for information (e.g. probability) on a normally-distributed variable.
5. Utilize the standard t-table to solve for information (e.g. probability) on a normally-distributed variable.
6. Describe and apply criteria determining if it is necessary to apply the Central Limit Theorem to solve for information (e.g. probability) on a normally-distributed variable.
7. Determine the impact of sample size on confidence intervals.